

Protection without Power: Tariffs, Exchange Rates, and the Determinants of Market Power and Misallocation in Indian Manufacturing

Laxmikanta Sarangi — Department of Humanities and Social Sciences, Indian Institute of Technology Roorkee **Falguni Pattanaik** — Department of Humanities and Social Sciences, Indian Institute of Technology Roorkee

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Abstract

We ask what moves market power and misallocation in Indian organised manufacturing. Using 423,127 plant-year observations from the Annual Survey of Industries (2012–13 to 2022–23), we estimate plant-level markups (De Loecker–Warzynski), markdowns and monopsony incidence, and within-industry TFPR dispersion (Hsieh–Klenow), and relate their industry- and state-level evolution to a newly constructed NIC-concorded applied-tariff panel (2011–2023), real-exchange-rate exposure, and state development. Three results stand out. First, protection does not buy market power: higher output tariffs are associated with *lower* markups in every specification (import-weighted tariffs: $p = 0.010$), the mirror image of the cost-channel result of De Loecker, Goldberg, Khandelwal and Pavcnik (2016) for liberalisation. Second, real appreciation compresses markups in import-exposed industries ($p = 0.038$ – 0.050), the classic import-discipline channel, while at finer industry detail protection is associated with lower monopsony incidence. A stable panel VAR confirms these effects dynamically and reveals political-economy feedback: markups and misallocation Granger-cause tariffs. Third, misallocation is orthogonal to trade policy but rises with state development ($p = 0.009$ excluding COVID years), consistent with transition dynamics rather than border distortions. Spatially, agglomerated states are uniformly more competitive and better allocated, and monopsony is concentrated in rural plants (44 versus 38 per cent incidence) in almost every major state.

JEL codes: D24, F13, F31, L11, L60, O14, R12 **Keywords:** markups; markdowns; monopsony; misallocation; tariffs; exchange rates; India; ASI

1. Introduction

Two decades after India’s landmark trade liberalisation, its trade policy has turned. Applied manufacturing tariffs, which fell steadily from the early 1990s to the mid-2010s, have risen since 2018; simultaneously, the rupee’s real effective exchange rate has appreciated persistently, and trade openness has fallen by nearly twenty percentage points of GDP from its 2012 peak. This paper asks what this reversal has done to the two central margins of allocative performance in Indian manufacturing: market power — in product markets (markups) and labour markets (markdowns) — and the misallocation of resources across plants.

The question inherits a rich literature but faces it with new data and a new episode. The liberalisation studies of the 1990s and 2000s found pro-competitive discipline (Krishna & Mitra, 1998), productivity gains (Topalova & Khandelwal, 2011), and — crucially — the cost-channel surprise of De Loecker, Goldberg, Khandelwal and Pavcnik (2016; hereafter DLGKP): tariff cuts *raised* markups, because input-cost declines were passed through to prices incompletely. Whether the protection episode of the late 2010s produced the mirror image, and whether it reached labour-market power and misallocation at all, is unknown. We provide, to our knowledge, the first plant-level evidence.

We build three panels from the ASI: industry $\times$ year (23 NIC-2008 divisions and 69 3-digit groups, 2012–2022) and state $\times$ year (36 states/UTs), carrying four outcome objects estimated at the plant level — the median DLW markup, the median markdown, monopsony incidence, and within-industry TFPR dispersion. We match them to a newly constructed tariff panel: India’s effectively-applied (AHS) tariffs at HS-6 from WITS, 2011–2023, concorded to NIC-2008 through the official UN HS CPC 2.1 ISIC Rev.4 correspondences — in the tradition of Debroy and Santhanam (1993) — validated line-by-line, and aggregated at both industry levels. Identification follows the exposure logic of the India trade literature: tariffs vary at industry $\times$ year;

the national REER enters interacted with pre-determined (2012) import exposure; state development enters state panels with two-way fixed effects throughout and clustering at the unit level. A panel VAR (Abrigo & Love, 2016), stable after removal of common time effects (Love & Zicchino, 2006), adds dynamics and feedback.

Our first and central result is that **protection did not buy market power**. The tariff coefficient on the median markup is negative in every one of ten specifications across two industry levels; with import-weighted tariffs it is -0.0028 ($p = 0.010$), excluding the COVID years -0.0050 ($p = 0.050$), with national controls -0.0063 ($p = 0.021$). A ten-point tariff increase is associated with a markup lower by 0.03–0.06. This is DLGKP’s economics running in reverse: tariffs raise the cost of imported inputs, and incomplete pass-through of those cost increases into prices compresses measured margins. The exchange rate tells the complementary story: **real appreciation compresses markups precisely in import-exposed industries** (-0.039 , $p = 0.038$ at NIC-2; -0.060 , $p = 0.050$ at NIC-3) — import discipline is alive, but it now operates through prices rather than tariff schedules.

Second, the labour market responds where the product market does not. At the 3-digit level, higher tariffs are associated with a *lower* incidence of monopsony ($p = 0.067$; robust to lags and controls), and appreciation with weakly higher monopsony in exposed industries. Protection props up product demand and, with it, competition for workers; import competition does the reverse. The panel VAR traces the same signs dynamically — a tariff shock lowers both markups and markdowns over one to four years — and adds a caution for identification everywhere in this literature: causality also runs backwards, with markups ($p = 0.016$) and misallocation ($p = 0.050$) Granger-causing tariffs. Indian protection in this decade responded to industry conditions.

Third, **misallocation is a domestic phenomenon**. Nothing in trade policy moves TFPR dispersion (no specification below $p = 0.38$); what moves it is state development, and in the direction opposite to the cross-country folklore: within states over time, higher real GSDP predicts *more* dispersion ($+39.1$, $p = 0.030$; $+44.4$, $p = 0.009$ excluding COVID). We read this through Asker, Collard-Wexler and De Loecker (2014): growth episodes draw in heterogeneous entrants and expand constrained plants unevenly, so measured dispersion widens along the transition path. The finding cautions against reading TFPR dispersion as a time-varying distortion index during growth accelerations.

Finally, geography. Employment is heavily agglomerated (top five states: 53 per cent), and the agglomeration gradient is uniformly pro-competitive: correlations of log state employment with markups (-0.30), markdowns (-0.28), monopsony (-0.26), misallocation (-0.22) and local labour-market concentration (-0.91). The sharpest spatial fact is rural–urban: rural plants are monopsonistic in 44.2 per cent of plant-years against 38.4 per cent urban — a gap present in every year and in almost every major industrial state (up to $+21$ percentage points in West Bengal and Uttar Pradesh), reversing only where rural employers are plantation-scale (Bihar, Assam). Labour-market power in India is a thin-market, rural phenomenon; product-market power is spatially uniform; misallocation is an eastern-states phenomenon.

The paper proceeds as follows. Section 2 situates the contribution. Section 3 describes the data and the construction of the tariff and exposure panels. Section 4 lays out the empirical strategy. Section 5 presents the static results, Section 6 the dynamics and robustness, Section 7 the spatial evidence, and Section 8 concludes.

2. Related Literature

The paper joins three literatures. The *trade-and-markups* literature for India runs from the pro-competitive evidence of Krishna and Mitra (1998) through Topalova (2010) and Topalova and Khandelwal (2011) to DLGKP (2016), whose input-cost channel our results extend to a protection episode; methodologically we follow the production-function markup tradition of De Loecker and Warzynski (2012) with the ACF control-function estimator. The *monopsony/markdown* literature (Brooks, Kaboski, Li & Qian, 2021, for India and China; Yeh, Macaluso & Hershbein, 2022, for the US) has documented pervasive labour-market power in manufacturing; our contribution is its first linkage to trade policy and exchange rates in a developing-country panel. The *misallocation* literature descends from Hsieh and Klenow (2009); our null result on trade policy and positive result on state growth speak to the debate on what TFPR dispersion measures (Asker et al.,

2014; Bils, Klenow & Ruane, 2021). On dynamics, we implement the panel VAR of Abrigo and Love (2016) with the time-demeaning of Love and Zicchino (2006); on political economy, the reverse Granger causality connects to the endogenous-protection tradition of Grossman and Helpman (1994) and, for India, Bown and Tovar (2011).

3. Data

Plant panel and outcome objects. The ASI unit-level data with common factory identifiers yield a cleaned panel of 423,127 plant-years (142,493 plants) over 2012–13 to 2022–23, deflated to 2011–12 prices (WPI by industry for output and materials; machinery WPI for capital; details in the thesis pipeline documentation). Markups are estimated by the DLW production-function approach with a translog gross-output technology per NIC-2 division (LP/ACF moments); markdowns follow Brooks et al. (2021), $\mu = (_L / _L) / (_M / _M)$, with $\mu > 1$ indicating monopsony; misallocation is the within NIC-4 $\times$ year variance of log TFPR at $\mu = 3$ (Hsieh–Klenow). Divisions 18 and 30 are excluded from markdown aggregates for data-quality reasons established in the thesis. Outcome panels take industry- or state-level medians (markup, markdown), the monopsony share, and the TFPR dispersion.

Tariffs. From WITS we extract India’s applied tariffs (AHS, with MFN fallback) at HS-6 for 2011–2023 — 85,793 line-years — and build an HS-6 $\rightarrow$ NIC-2008 concordance through the official UN correspondence tables (HS2012 CPC 2.1, HS2017 CPC 2.1, CPC 2.1 ISIC Rev. 4), with an HS-4 heading fallback for HS-2022-vintage lines; 4,847 manufacturing HS-6 codes map, and the concordance validates against every HS chapter’s expected division. The headline industry tariff is the simple average across HS-6 lines (Topalova convention; import-weighted as robustness), giving complete panels at NIC-2 (23×13) and NIC-3 (68×13). The 2022 structure — beverages 110 per cent, motor vehicles 40, food 28, tobacco 33, against metals 5.5, electronics 6.2, chemicals 6.6 — matches India’s known protection profile.

Exchange rates, exposure, and state accounts. The REER/NEER are the BIS broad indices averaged to fiscal years. Import exposure is the industry’s 2012 share of manufacturing imports (pre-determined; frozen at base year). State real GSDP (constant 2011–12) and per-capita NSDP come from the MOSPI state accounts (March-2024 vintage), matched to ASI state codes; local labour-market concentration is the employment HHI of plants within state $\times$ NIC-3 $\times$ year cells, aggregated employment-weighted.

4. Empirical Strategy

The static model is a two-way fixed-effects exposure design,

$$Y_{it} = _it + (\text{REER}_t - 100) \times s_i^{\text{imp}} + _i + _t + _it,$$

for industry panels, with $Y \in \{\text{median markup, median markdown, monopsony share, TFPR dispersion}\}$, and $Y_{st} = \ln \text{GSDP}_{st} + \text{HHI}_{st} + _s + _t + _st$ for states. Year effects absorb all national aggregates; identification of $_it$ is shift-share, from pre-determined exposure; standard errors cluster at the unit level. Robustness: one-year lags; import-weighted tariffs; excluding 2020–21; replacing year effects with observed national controls (REER, GDP growth, CPI inflation); GSDP growth and per-capita variants at the state level — 72 coefficient estimates in all.

Dynamics use a panel VAR in $(_, _, _, V)$, GMM with forward orthogonal deviations, instruments lags 1–3, lag order 1 by MBIC. The levels system is explosive (maximum eigenvalue modulus 1.107) because of common secular trends; year-demeaning (Love & Zicchino, 2006) restores stability (0.713 at NIC-3; 0.740 at NIC-2). Orthogonalised IRFs order the tariff first; 200 Monte Carlo draws give the bands.

5. Results: Trade Policy, the Exchange Rate, and Market Power

Table 1 presents the core estimates (industry panels; state panel in column 5).

Table 1 — Determinants of market power and misallocation (two-way FE)

Outcome / regressor	NIC-2	NIC-3	NIC-2 robust range	NIC-3 monopsony	State
Markup $\leftarrow$ tariff	−0.0048* (0.086)	−0.0021 (0.206)	−0.0028** to −0.0063**	—	—
Markup $\leftarrow$ REER $\times$ exposure	−0.0391** (0.038)	−0.0595** (0.050)	—	—	—
Markdown $\leftarrow$ tariff	0.0009 (0.757)	−0.0012 (0.748)	n.s.	—	—
Monopsony $\leftarrow$ tariff	0.0003 (0.853)	−0.0019* (0.067)	—	−0.0018* to −0.0010	—
Misallocation $\leftarrow$ tariff	−0.0001 (0.999)	0.0014 (0.988)	n.s.	—	—
Misallocation $\leftarrow$ $\ln$ GSDP	—	—	—	—	+39.1** (0.030)
Markdown $\leftarrow$ labour HHI	—	—	—	—	−0.44* (0.083)

Notes. p-values in parentheses, clustered by unit. N = 253 (NIC-2), 748 (NIC-3), 350 (state). “Robust range” spans lag, import-weighted, no-COVID and national-control specifications; all are negative for the markup–tariff pair, significant at 1–5 per cent in three of five.

Three facts. (i) The markup–tariff association is negative always and everywhere — never positive in any of the ten industry specifications. Interpreted through DLGKP’s cost channel, higher tariffs raise material input costs (intermediates dominate India’s imports); with incomplete pass-through, measured price–cost margins compress. The point estimates imply that the post-2018 tariff increase (roughly +2 points on the manufacturing mean) is associated with a markup lower by about 0.01, or close to one per cent of the median markup level. (ii) The REER-exposure coefficient is negative, significant, and larger at the finer industry level — appreciation disciplines exposed industries’ margins. (iii) Trade policy does not touch TFPR dispersion at any level or in any specification, but state development does, positively — a result to which we return in Section 7.

6. Dynamics, Feedback, and Robustness

The stable panel VAR confirms the static signs dynamically: an orthogonalised tariff shock lowers the markup for one to four years and the markdown for one to two, and temporarily raises misallocation — a short-lived allocative disruption from discrete protection episodes consistent with the Granger p-value of 0.081. The bidirectional structure is equally instructive. Markups Granger-cause tariffs ($p = 0.016$) and misallocation Granger-causes tariffs ($p = 0.050$): protection responded to industry conditions — rising where margins had fallen and dispersion had grown — placing India’s recent episode squarely in the endogenous-protection tradition and reinforcing our reliance on pre-determined exposure shares for the static identification. At the 2-digit level, the markup and markdown Granger-cause each other ($p = 0.001, 0.009$), the industry-frequency reflection of the plant-level joint determination of the two margins documented in the thesis’s market-power chapter.

Across the 72-specification robustness grid, the qualitative pattern never flips: the markup–tariff coefficient is negative in 10 of 10 industry cells; the REER-exposure effect is negative on markups in all variants; the misallocation–trade nulls are universal; and the state-GSDP–misallocation coefficient is positive in all five variants, strongest excluding COVID (+44.4, $p = 0.009$) and in per-capita terms (+37.7, $p = 0.065$), and absent in short-run growth rates — a level, not a cycle, effect.

7. Spatial Evidence

Manufacturing employment is agglomerated (state employment HHI 0.077; top five states 53 per cent) and industries differ enormously in spatial concentration (tobacco 0.565; leather 0.215; apparel 0.203; chemicals

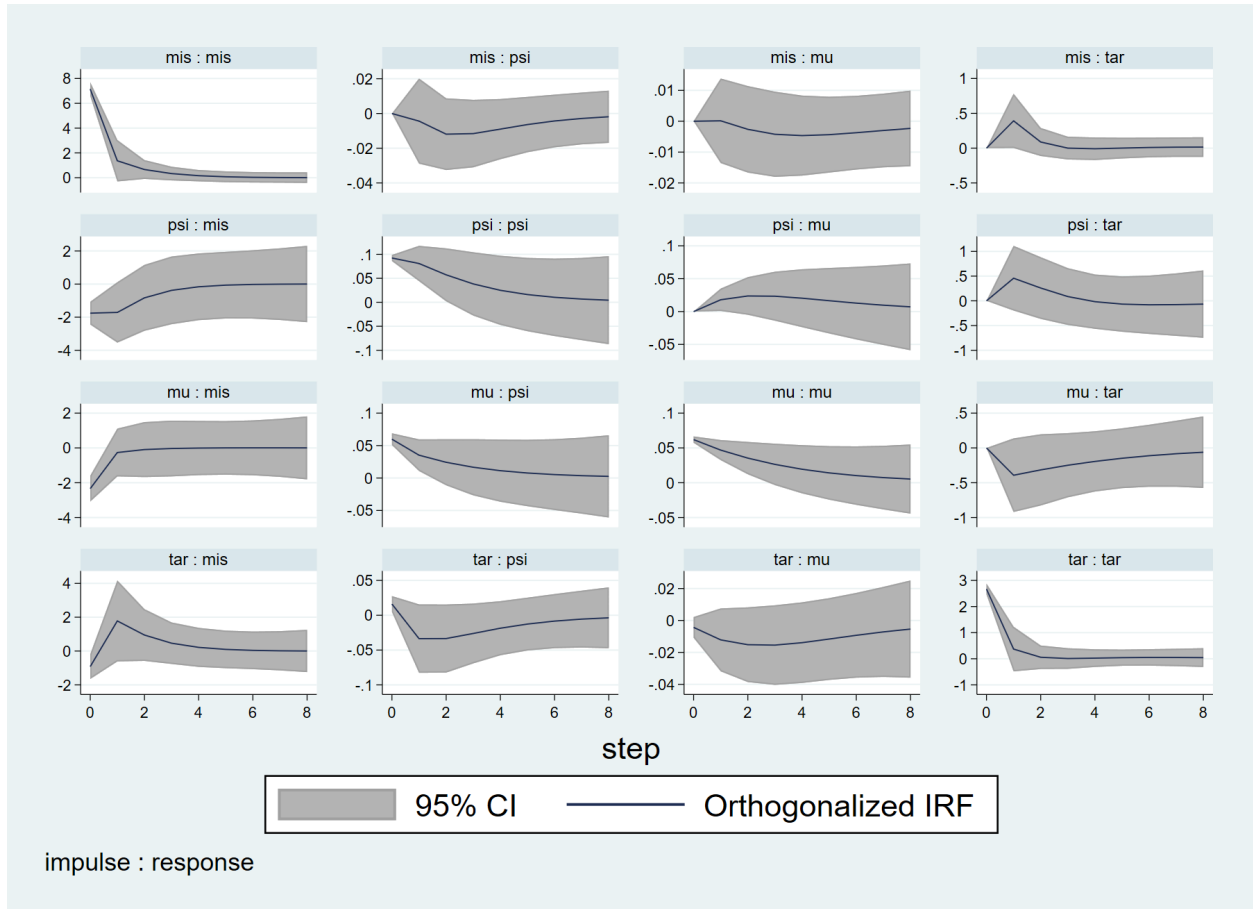


Figure 1: Figure 3 — Orthogonalised impulse responses, year-demeaned NIC-3 panel VAR (95 per cent bands, 200 Monte-Carlo draws).

0.128). The plant-weighted agglomeration gradient is uniformly pro-competitive: state size correlates at -0.30 with the median markup, -0.28 with the markdown, -0.26 with monopsony incidence, -0.22 with misallocation and -0.91 with local labour-market concentration. Misallocation is geographically eastern (Odisha 77.6, Andhra Pradesh 76.5, West Bengal 75.9, against Punjab 49, Gujarat 56.5); monopsony peaks in the central-eastern mineral belt (Chhattisgarh 0.65, Odisha and Jharkhand 0.55).

Median markup by state, 2012-2022

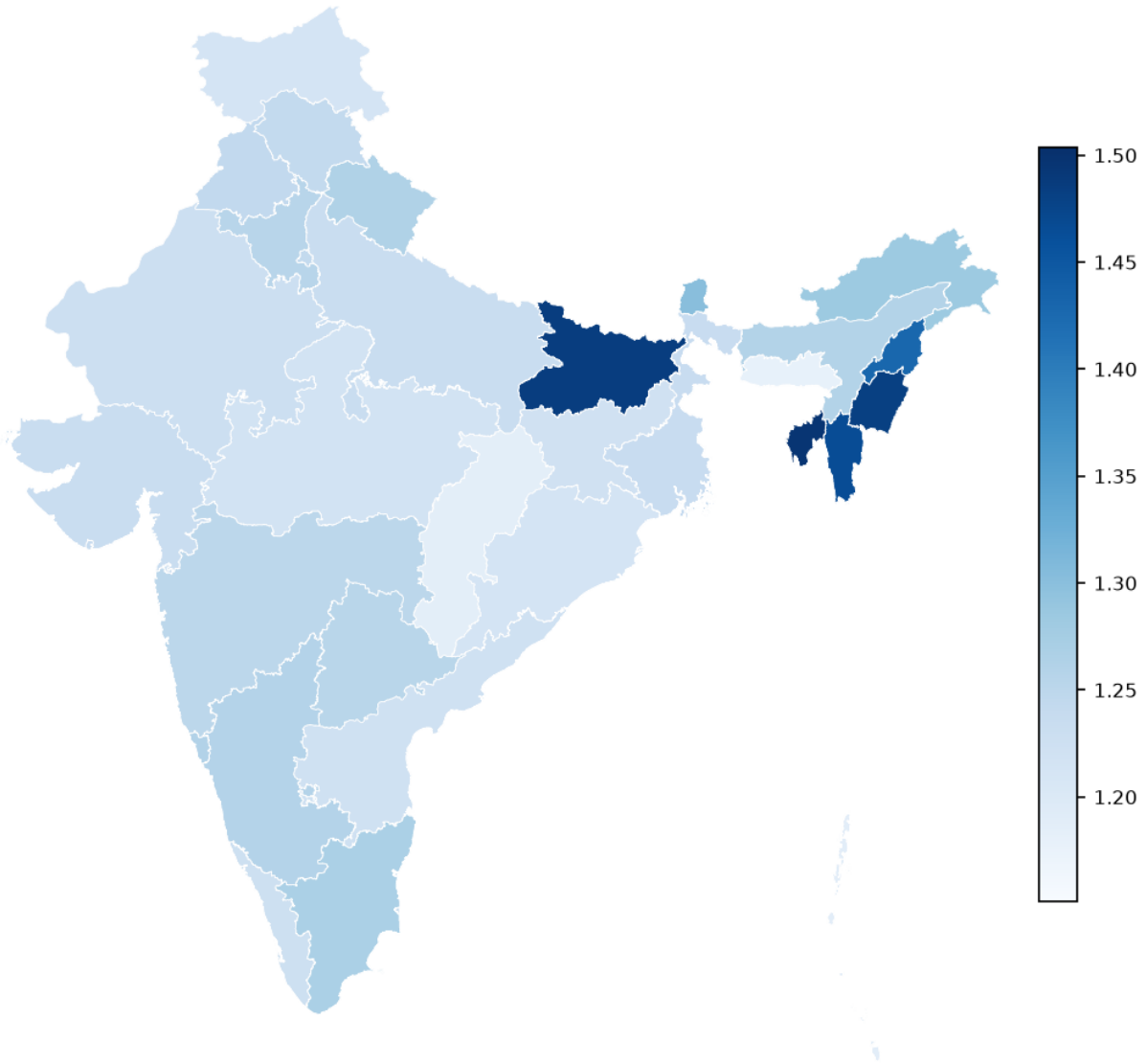
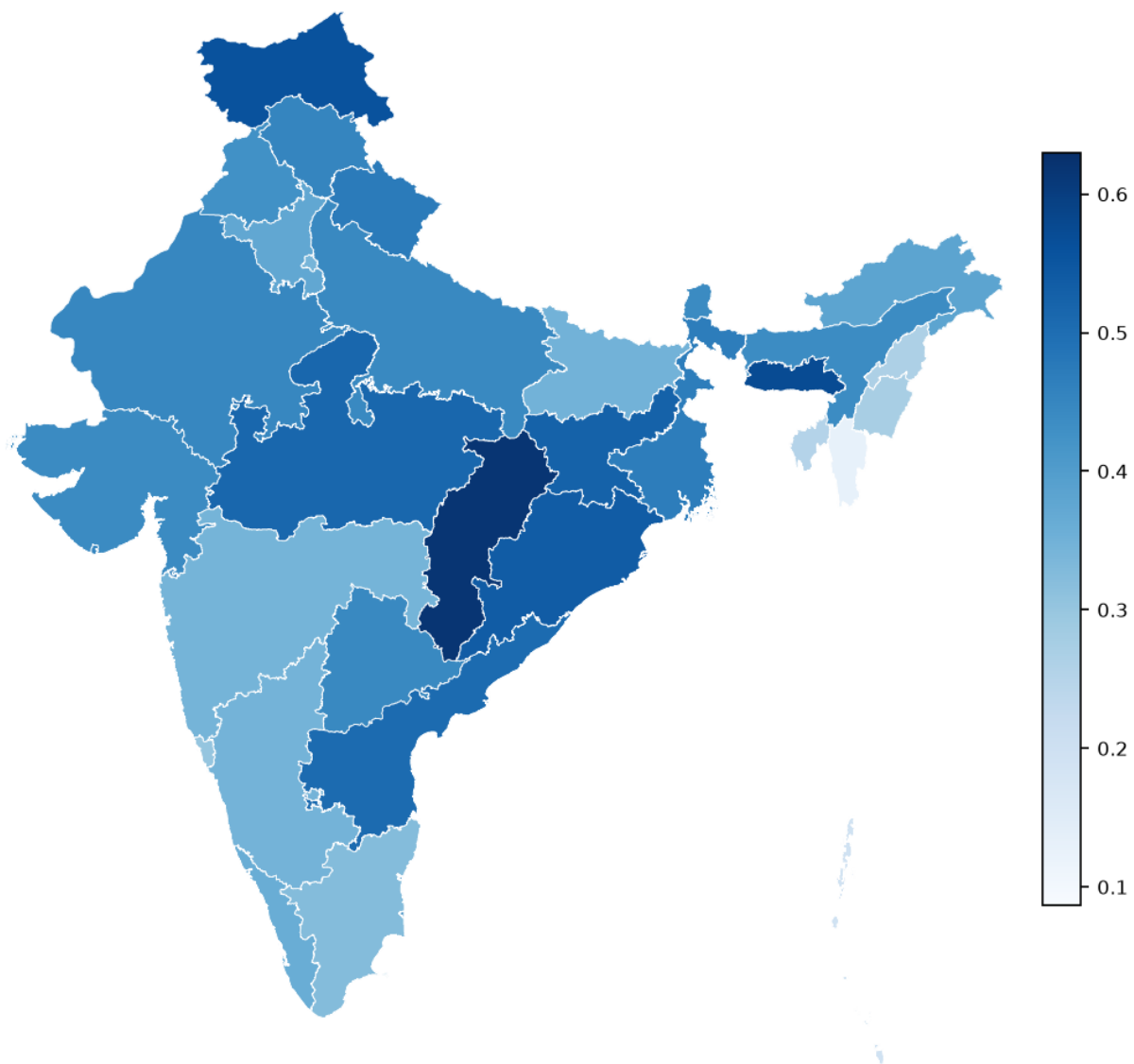
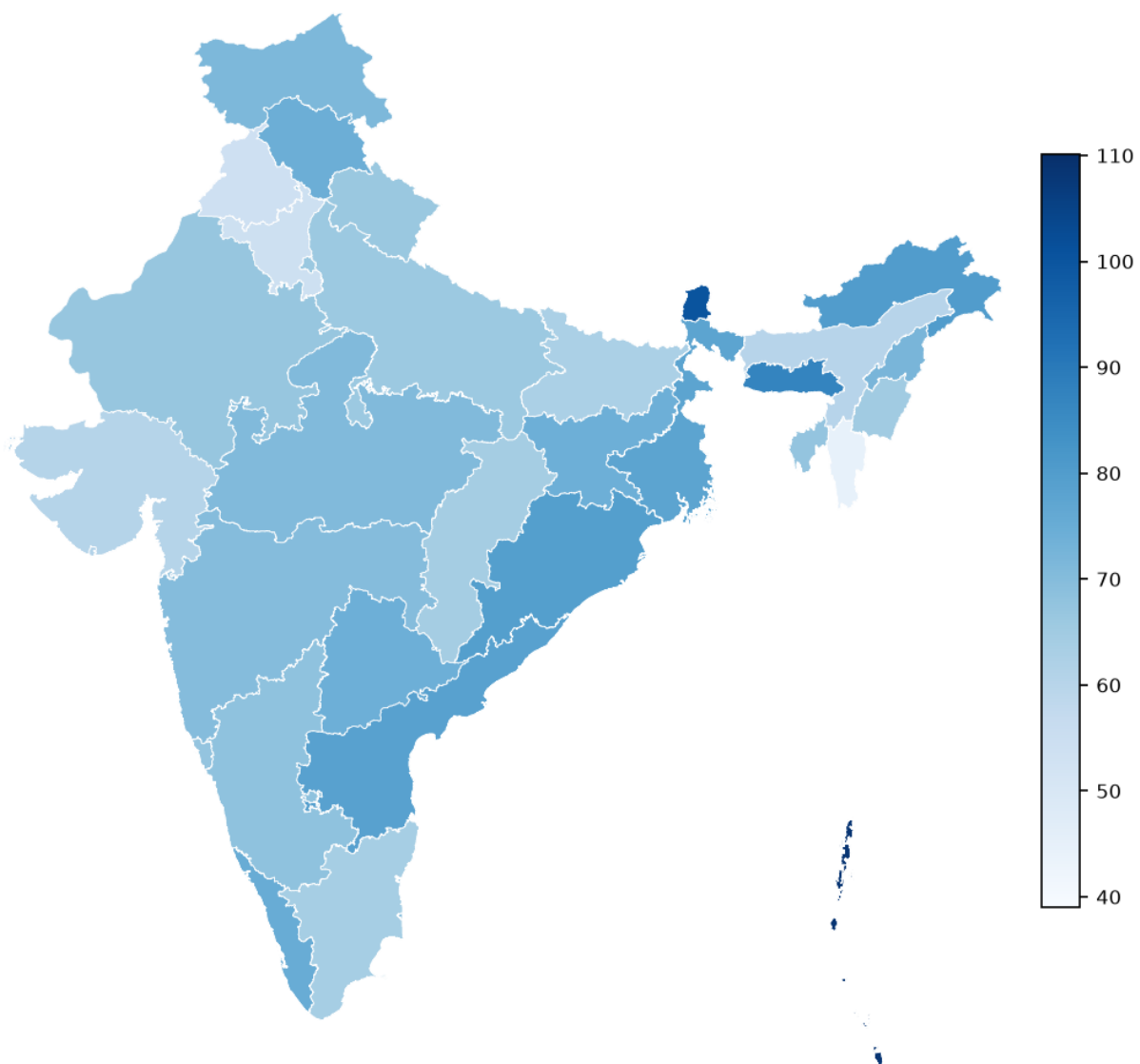


Figure 2: Figure 1 — The geography of Indian manufacturing market power and misallocation, 2012–2022: (a) median markup, (b) monopsony incidence, (c) misallocation by state.

Monopsony incidence by state (share with $\text{psi} > 1$), 2012-2022

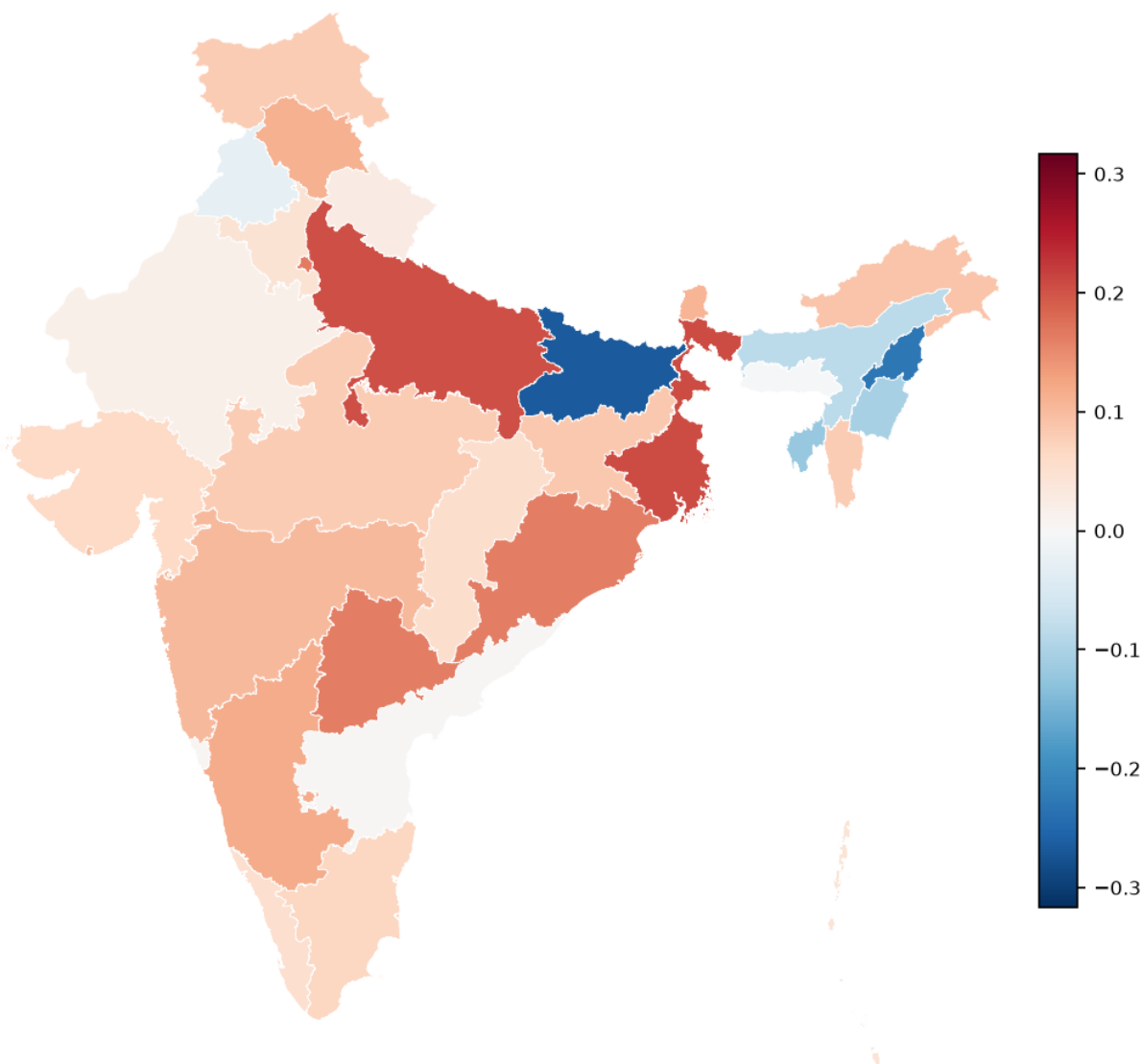


Misallocation by state (var log TFPR x100), 2012-2022



Within states, the rural–urban split isolates where labour-market power lives. Rural plants (44 per cent of plant-years) are monopsonistic in 44.2 per cent of cases versus 38.4 urban, with near-identical markups (1.217 vs 1.233) and misallocation (64.4 vs 63.1): the sectors differ *only* on the labour-market margin. The gap holds in every sample year and in most major states — West Bengal +20.8 points, Uttar Pradesh +20.5, Odisha +16.0, Telangana +15.8, Karnataka +11.0, Maharashtra +9.3 — and reverses precisely where the mechanism predicts: Bihar (−31.7), Assam (−11.7) and Punjab (−5.4), states whose rural manufacturing is dominated by large agro-processing employers facing thin urban sectors.

Rural minus urban monopsony incidence (pp gap), 2012-2022



Local labour-market concentration itself declined all decade (employment-weighted HHI 0.374 $\rightarrow$ 0.343; mean plants per market 24.3 $\rightarrow$ 29.5), consistent with the secular decline in monopsony incidence.

8. Conclusion

India's protection turn of the late 2010s did not restore manufacturing market power; on the contrary, tariffs are associated with lower markups — the cost channel of De Loecker et al. (2016) in reverse — while the appreciating rupee disciplined margins in import-exposed industries. Labour-market power responded on the opposite margin, falling with protection at fine industry detail. Misallocation proved beyond the reach of trade policy entirely, rising instead with state development along the transition path. And the geography is stark: product-market power is spatially flat, but monopsony is a rural, thin-market phenomenon and misallocation an eastern-states one. For policy, the results imply that tariffs are a poor instrument for either competition or allocative goals; that exchange-rate appreciation carries a quiet pro-competitive dividend; and that the binding margins — rural labour-market power and eastern-state misallocation — call for domestic market-thickening and factor-market reform, not border instruments.

References

- Abrigo, M. R. M., & Love, I. (2016). Estimation of panel vector autoregression in Stata. *The Stata Journal*, 16(3), 778–804.
- Asker, J., Collard-Wexler, A., & De Loecker, J. (2014). Dynamic inputs and resource (mis)allocation. *Journal of Political Economy*, 122(5), 1013–1063.
- Bils, M., Klenow, P. J., & Ruane, C. (2021). Misallocation or mismeasurement? *Journal of Monetary Economics*, 124, S39–S56.
- Bown, C. P., & Tovar, P. (2011). Trade liberalization, antidumping, and safeguards: Evidence from India’s tariff reform. *Journal of Development Economics*, 96(1), 115–125.
- Brooks, W. J., Kaboski, J. P., Li, Y. A., & Qian, W. (2021). Exploitation of labor? Classical monopsony power and labor’s share. *Journal of Development Economics*, 150, 102627.
- Debroy, B., & Santhanam, A. T. (1993). Matching trade codes with industrial codes. *Foreign Trade Bulletin*, 24(1), 5–27.
- De Loecker, J., Goldberg, P. K., Khandelwal, A. K., & Pavcnik, N. (2016). Prices, markups, and trade reform. *Econometrica*, 84(2), 445–510.
- De Loecker, J., & Warzynski, F. (2012). Markups and firm-level export status. *American Economic Review*, 102(6), 2437–2471.
- Grossman, G. M., & Helpman, E. (1994). Protection for sale. *American Economic Review*, 84(4), 833–850.
- Hsieh, C.-T., & Klenow, P. J. (2009). Misallocation and manufacturing TFP in China and India. *Quarterly Journal of Economics*, 124(4), 1403–1448.
- Krishna, P., & Mitra, D. (1998). Trade liberalization, market discipline and productivity growth: New evidence from India. *Journal of Development Economics*, 56(2), 447–462.
- Love, I., & Zicchino, L. (2006). Financial development and dynamic investment behavior: Evidence from panel VAR. *The Quarterly Review of Economics and Finance*, 46(2), 190–210.
- Topalova, P. (2010). Factor immobility and regional impacts of trade liberalization: Evidence on poverty from India. *American Economic Journal: Applied Economics*, 2(4), 1–41.
- Topalova, P., & Khandelwal, A. (2011). Trade liberalization and firm productivity: The case of India. *Review of Economics and Statistics*, 93(3), 995–1009.
- Yeh, C., Macaluso, C., & Hershbein, B. (2022). Monopsony in the US labor market. *American Economic Review*, 112(7), 2099–2138.